

Action–Evidence Geometry and Execution Certificates

Supplementary Evidence

Anonymous submission

This supplement reports the complete benchmark tables, the matched geometry controls, and the reproducibility details for the deterministic DAFR execution layer. The main paper’s claims are limited to typed action–evidence encoding, joint-risk execution constraints, and certificate-based diagnosis and repair. The matched studies do not claim that geometric code is more expressive or intrinsically safer than unrestricted predicates.

Evaluation Protocol

AgentDojo. AgentDojo v1.2.2 contains Banking, Slack, Travel, and Workspace. We evaluate DeepSeek V4 Flash, GPT-5.4-mini, and Claude Haiku 4.5 on 71 clean and 585 adversarial cases. The clean/attack counts are 16/144, 21/105, 20/140, and 14/196 for the four suites. Clean utility, attack utility, and ASR are unweighted macro-averages over suites.

Agent Security Benchmark. The ASB comparison contains 204 cases with DeepSeek V4 Flash. It reports utility, task success, and ASR. The reported DAFR run uses the benchmark’s source-tagged runtime configuration; the source setting is part of the threat model and should not be generalized to settings in which provenance is unavailable.

Aggregation and result provenance. The main AgentDojo protocol uses four suites (71 clean and 585 attacked episodes). The archived three-suite subset contains 57 clean and 389 attacked episodes; its 88.1% clean and 83.2% attack utility for DAFR are different aggregates, not alternative estimates for the four-suite table. For metric x , suite macro-averaging is $\bar{x} = \frac{1}{4} \sum_{s=1}^4 x_s$, rather than pooling all episodes. The four-suite DAFR aggregate is 87.5% clean and 81.4% attack utility.

The retained GPT baseline table combines earlier three-suite document aggregates with Workspace reruns; the DAFR row is recorded as a four-suite rerun. These baseline rows must not be described as newly reproduced end-to-end runs. The plotted multi-model tables retain archived aggregates; paired statistical comparisons require the underlying episode outcomes. The ASB aggregate specifies seed

17, stratified attack sampling, round-robin case sampling, shared workflow plans, and source-tagged tool provenance. Its task-success score checks completion of required normal tools, while ASR is the stored attack-goal match or an executed attacker-tool trace. These protocol choices limit direct comparison with other ASB settings.

Execution boundary. DAFR runs after a planner proposes a concrete call and before the trusted runtime executes it. The deterministic typed field interface validates registered fields and binds them to shared effect roles. The action–evidence encoder retains field identity, evidence source, authorization, and state relations. No language model is used by the execution verifier or by the matched geometry controls.

Matched Geometry and Certificate Studies

The membership arms share the same risk vectors; the repair arms share the same typed actions, evidence, and candidate transformations. Axis-ifelse checks each risk coordinate independently. Geometry evaluates one shared joint-risk region. The equivalent predicate computes the same halfspace and norm inequalities with ordinary arithmetic and is included as an implementation-equivalence control. A Boolean-only interface has no repair operator. Its zero-repair entry in the archived runner is fixed by construction and is excluded from empirical comparisons here.

The joint-risk suite contains 512 four-dimensional vectors. Every coordinate is below its independent 0.8 threshold, while 25 vectors exceed the shared joint budget. Geometry and the equivalent predicate reject the same 25 vectors. The repair suite contains 256 local write cases: 192 have a safe candidate obtained by removing an unsupported optional destination, and 64 require blocking because the destination is semantically required. Geometry and a predicate supplied with the same repair oracle recover the same 192 actions. The equality is expected because the same executable repair transformation is used in both arms; it does not show an intrinsic geometric repair advantage.

We additionally audited 614 saved AgentDojo decision records. Four records were malformed and 610 were well formed. Rechecking the recorded hard margins with an equivalent slack predicate agreed on 610/610 records; 546 were allowed and 64 were blocked or clarified. This is a saved-certificate consistency audit, not a new planner replay, exter-

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Backend	Safe allow	Unsafe block	False allow
Axis thresholds	487/487	0/25	25/25
Joint geometry	487/487	25/25	0/25
Equivalent predicate	487/487	25/25	0/25

Table 1: Membership under a specified unit-norm risk policy: 487 feasible and 25 infeasible vectors. All arms have zero false blocks. Labels are determined analytically by that policy, rather than independent real-world safety annotations.

Backend + shared repair	Repaired	Required blocks
Geometry	192/192	64/64
Equivalent predicate	192/192	64/64

Table 2: Local action-repair audit: both backends block all 256 original proposals and preserve the intended effect in all 192 repairable cases (192/256 overall). The same transformation and effect check are used in both arms. A decision-only interface has no repair operator and is not treated as an empirical repair competitor.

nal execution, or utility/ASR measurement.

Fixed-family stress test. A 64-case in-process suite crosses fresh and stale state with four fixed injection families: forged approval or runtime metadata, policy-override instructions, Chinese webpage instructions, and fabricated prerequisite claims. The runtime blocks 64/64 injected writes, including 32/32 fresh-state and 32/32 stale-state cases. An unprotected local executor performs 64/64 writes. The suite has no external side effects and is reported as a mechanism stress test rather than an adaptive-attack result.

Complete AgentDojo Results

Values are percentages. Clean and attack utility are higher-is-better; ASR is lower-is-better. The full case counts are given in the protocol above.

Method	Clean	Attack	ASR
No Defense	86.8	75.4	17.7
Sandwich	89.8	76.2	12.0
Prompt	85.0	79.2	3.9
Spotlighting	74.4	77.8	11.3
PIGuard	44.9	42.5	0.0
ProtectAI	48.0	36.2	4.2
Progent	75.2	71.7	2.1
DRIFT	62.6	53.2	1.4
DAFR	87.5	81.4	0.0

Table 3: AgentDojo results with DeepSeek V4 Flash.

Across the three planners, DAFR records 0 observed ASR on the evaluated static attacks. Relative to PIGuard, the other method with 0 observed ASR in all three planner settings, the attack-utility differences are 38.9, 25.1, and 30.2 percentage points. These are descriptive differences on the fixed benchmark cases, not claims of statistical superiority.

Method	Clean	Attack	ASR
No Defense	72.2	61.2	5.2
Sandwich	80.6	65.4	2.1
Prompt	76.6	65.4	0.6
Spotlighting	70.5	62.7	2.5
PIGuard	46.3	34.4	0.0
ProtectAI	48.9	38.6	2.3
Progent	66.0	54.2	1.4
DRIFT	41.7	38.8	0.2
DAFR	68.7	59.5	0.0

Table 4: AgentDojo results with GPT-5.4-mini.

Method	Clean	Attack	ASR
No Defense	74.1	66.3	0.2
Sandwich	75.4	67.1	0.0
Prompt	70.9	65.1	0.0
Spotlighting	74.1	62.5	0.0
PIGuard	51.6	38.2	0.0
ProtectAI	56.0	43.9	2.1
Progent	65.3	58.1	0.0
DRIFT	43.1	44.1	0.2
DAFR	70.4	68.4	0.0

Table 5: AgentDojo results with Claude Haiku 4.5.

Agent Security Benchmark

Method	Utility	Task success	ASR
No Defense	88.7	77.9	90.2
Delimiter	87.3	77.0	86.3
Instruction Prevention	88.0	77.5	39.7
Observation Sandwich	88.0	77.0	27.9
TS-Flow	85.5	73.5	19.1
Progent	86.0	73.0	3.4
DRIFT	85.8	73.0	23.0
DAFR	86.8	73.5	0.0

Table 6: ASB results on 204 cases with DeepSeek V4 Flash. DAFR’s ASR is 0/204 observed attacks.

System Ablation and Signed Margins

The system ablation uses the same 71 clean and 585 attacked cases with DeepSeek V4 Flash. These rows remove interacting execution modules and therefore do not isolate the syntax of the geometric boundary. The matched representation study above supplies that narrower comparison.

The admitted/rejected sign separation is a consistency check for the decision rule. It is not a calibration or security theorem. The local matched suite and saved-certificate audit are therefore reported separately from benchmark utility and ASR.

Reproducibility Record

The evaluated configuration uses seven shared effect roles, 45 action-evidence coordinates, 13 hard constraint templates,

Configuration	Clean	Attack	ASR
DAFR	87.5	81.4	0.0
Without evidence projection	79.2	81.8	0.0
Without action–evidence encoding	91.5	85.3	0.7
Without dynamic defense path	90.2	81.3	16.7
Without decision feedback	79.4	80.0	0.8

Table 7: System-level ablation on AgentDojo with DeepSeek V4 Flash.

Setting	Calls	Admitted mean	Rejected mean	Gap
AgentDojo / DeepSeek	3947	0.712	−0.368	1.080
AgentDojo / GPT	2916	0.748	−0.214	0.963
AgentDojo / Claude	2220	0.750	−0.289	1.039
ASB / DeepSeek	488	0.134	−0.885	1.018

Table 8: Signed margins; M_t is the minimum hard-constraint slack.

and one soft confidence facet. Coefficients and risk budgets are configuration constants; freezing them before scoring does not substitute for a parameter-sensitivity analysis. Every decision records the concrete call, selected constraints, margin vector, source links, and final decision. The benchmark comparisons use fixed case sets and report malformed or blocked calls through the benchmark’s native scoring functions. No matched geometry case invokes a real external tool.

The repository contains the benchmark manifests, raw decision records, aggregation scripts, and the matched geometry runner. The local suites should be interpreted as mechanism tests: they establish coupled-risk and certificate behavior under controlled inputs, while AgentDojo and ASB provide the task-level utility and observed attack outcomes.